

Manoj Kumar V

[Portfolio](#) — [GitHub](#) — [LinkedIn](#)

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Experience

Associate Software Engineer – Backend & Cloud(1.5 YOY)

Mar 2025 - Present

Texas AI Pvt. Ltd., Bangalore

Project 1: Owl Intelligence Platform (Law Enforcement & Investigation):

- Architected a cross-cutting security filter executing immediately after JWT authentication across all microservices; successfully intercepting and evaluating dynamic permissions for **5,000+ daily requests** with zero latency overhead.
- Scaled data pipelines for **2TB concurrent file streams** using S3 multi-part chunking and pessimistic locking, eliminating race conditions and ensuring **100% transactional reliability**.
- **Automated serverless** DevSecOps workflows for **500+ daily massive uploads** using EventBridge and Lambda for GuardDuty malware scans, cutting vulnerability response times by **40%**.
- **Orchestrated a production GenAI gateway** by migrating auto-scaled EC2 Ollama clusters to Bedrock Agents with a token billing engine, reducing model inference latency by **30%** and infrastructure overhead.

Project 2: Heita (Health Insurance SaaS Platform):

- Automated medical data extraction from handwritten files and insurance cards using FastAPI and MongoDB, accelerating structural query speeds into the Availity Clearinghouse API by 35%.
- **Designed and trained a Conversational AI chatbot** from scratch using the **RASA framework**, automating 60%+ of primary customer support intents and reducing user drop-off metrics.
- **Engineered the core monetization layer** by independently integrating the **Stripe API**, managing multi-currency webhook lifecycles with built-in retry logic for **100% financial transaction reliability**.
- **Streamlined continuous deployment (CI/CD)** and configuration scaling on distributed EC2 environments, cutting deployment times by **50%** and maintaining application uptime.

Projects

[MindEase](#) — AI Mental Wellness Tracker | *FastAPI, React, DynamoDB, Bedrock, CloudFront*

[GitHub](#)

- **Architected a full-stack cloud-native app** with DynamoDB, optimizing data paths and accelerating feature delivery by **25%**.
- Designed a dual-auth model supporting JWT and cryptographic keys, ensuring 100% tenant isolation and 24-hour session sandboxing.
- **Implemented multi-turn GenAI chat** and journal insights using Bedrock, cutting inference latency by **35%** via strict JSON parsing.
- **Optimized global asset delivery** by hosting the frontend on S3 via CloudFront CDN, reducing initial page load times by **40%**.

Student's Stress Level Prediction | *Spring Boot, Python, KNN, Naive Bayes, Lambda*

[GitHub](#)

- Linked Java backend with Python ML models (KNN & Naive Bayes) using serialized JSON streams, reducing inference latency by **40%**.
- **Engineered serverless data-retrieval workflows** using Lambda and RAG architectures for dynamic semantic context extraction.
- **Authored LLD specifications** and integrated cross-platform log analytics, minimizing runtime errors across system boundaries.

Technical Skills

Languages: Java, Python

Backend: Spring Boot, Spring MVC, FastAPI, REST APIs, Webhooks, Microservices

Messaging and Infrastructure: Apache Kafka, SNS, SES, Celery

Databases & Caching: MySQL, MongoDB, Redis(Distributed Locking)

Cloud & DevSecOps (AWS): EC2, S3, EventBridge, Lambda, GuardDuty, Bedrock, Auto Scaling, IAM, CloudWatch, Jenkins

Concepts: DSA, OOP, LLD, System Design, CI/CD, Multithreading

Tools: Git, Jira, Bitbucket, Postman, Linux

Achievements

- **AWS Certified Developer – Associate** ([Badge](#))
- Led the personal project “Student’s Stress level Prediction” which was selected for the State-Level Exhibition by the **Karnataka State Council for Science and Technology** ([KSCST](#)).
- Ranked 1st Place in 'Level Up' Hackathon – ME-RIISE, Malnad College of Engineering

Education

Malnad College of Engineering, Hassan

2021 – 2025

B.E. in Information science and engineering

CGPA: 9.4

BGS PU College, Mysuru

2020

(Class XII)

Percentage: 94.5%